

Ordinary Differential Equations

Mid-Semester Exam

Total Points: 30

Time: 90 minutes

Important

1. Simplify your answers as much as possible and write them in terms of fractions or constants such as $\sqrt{e}$ or $\log(4)$ rather than decimals.
2. Show all your work and explain your reasonings clearly.
3. No books, notes, calculators or any papers are allowed. Additional paper is available upon request.
4. You are not allowed to communicate with other students during the exam. Copying will not be tolerated.

1. (10 points) Find the general solution of the following ODEs.

✓(a) $(e^{2t} - t^2 \sin x) \frac{dx}{dt} + 2xe^{2t} + 2t \cos x = 0.$ (3 points)

(b) $t \frac{dx}{dt} = x + \sqrt{x^2 + t^2}.$ (3 points)

✓(c) $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 2x = e^t.$ (4 points)

- ✓2. (4 points) Solve: $\frac{dx}{dt} = \frac{t}{1+t}x + 1$, with the initial condition: $x(0) = 0.$

- ✓3. (6 points) Let $\alpha, \beta > 0$ and consider:

$$x' = \beta x - \alpha x^2.$$

- ✓(a) Find three distinct solutions of the above equation.

- ✓(b) For every solution $x(t)$ of the above equation (obtained in (a)) with $x(0) > 0$ show that $x(t) > 0$ for all $t \geq 0$ and $\lim_{t \rightarrow +\infty} x(t) = \frac{\beta}{\alpha}.$

- ✓4. (4 points) Construct three distinct solutions of the the initial value problem: $\begin{cases} \dot{x} = |x|^{3/5}, \\ x(0) = 0. \end{cases}$

- ✓5. (3 points) Suppose $x(t)$ satisfies the differential equation: $2\dot{x} = (2\cos(2x) + 3)x.$ Using the Gronwall's inequality or otherwise show that

$$x^2(t) \leq x^2(0)e^{5t} \text{ for all } t \geq 0.$$

6. (3 points) Determine the values of the parameters $\alpha, \beta \in \mathbb{R}$ such that the linear system

$$X'(t) = \begin{pmatrix} \alpha & -\beta \\ \beta & 2 \end{pmatrix} X(t) \text{ has a sink at the origin.}$$

Ordinary Differential Equations

End-Semester Exam

Total Points: 50
Time: 150 minutes

Important

1. Simplify your answers as much as possible and write them in terms of fractions or constants such as $\sqrt{e}$ or $\log(4)$ rather than decimals.
2. Show all your work and explain your reasonings clearly.
3. No books, notes, calculators or any papers are allowed. Additional paper is available upon request.
4. You are not allowed to communicate with other students during the exam. Copying will not be tolerated.

1. (10 points) Solve the following:

(a) Find the general solution $x(t)$ of: $x' = \frac{2x(x-1)}{t(2-x)}$.

(b) Solve the initial value problem: $x'(t) = \cos(tx) - 1$, with $x(0) = 0$.

(c) Find the general solution $X(t)$ of $X'(t) = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix} X(t) + \begin{pmatrix} 0 \\ 5t \end{pmatrix}$.

2. (10 points) Show that there exists a unique function $x = x(t)$ defined for all $t > 0$ satisfying the following:

(a) $x'(t) = \begin{cases} 5 - 2x & \text{if } x < 0 \\ 3 & \text{if } x \geq 0, \end{cases}$ with $x(0) = 0$.

(b) $x'(t) = \frac{x e^{-t^2}}{1 + 2t^2 + x^4}$, with $x(0) = 0$.

(c) $x'(t) = \sum_{k=1}^{2024} (1 + \cos^2 x + \sin^4 x)^k$, with $x(0) = 0$.

3. (3 points) Let f and g be continuous functions on $[a, b]$ and suppose ϕ_1 and ϕ_2 satisfies:

$$x''(t) + f(t)x'(t) + g(t)x(t) = 0 \text{ for } t \in (a, b).$$

Assume ϕ_1 attains its maximum and ϕ_2 attains its minimum at some $\xi_0 \in (a, b)$. Then calculate the Wronskian $W(\phi_1, \phi_2)(t)$ for all $t \in (a, b)$, where

$$W(\phi_1, \phi_2)(t) := \begin{vmatrix} \phi_1(t) & \phi_2(t) \\ \phi_1'(t) & \phi_2'(t) \end{vmatrix}.$$

4. (3 points) Let $G: \mathbb{R} \rightarrow \mathbb{R}^2$ be a continuous function such that $\lim_{t \rightarrow +\infty} \|G(t)\| = 0$ and let A be a 2×2 matrix with real entries such that

$$\max\{\operatorname{Re}(\lambda) : \lambda \text{ is an eigenvalue of } A\} := \sigma < 0.$$

Consider $\phi(t, X_0)$ satisfying the initial value problem:

$$X'(t) = AX(t) + G(t), \quad X(0) = X_0 \in \mathbb{R}^2.$$

For any $X_0 \in \mathbb{R}^2$ calculate $\lim_{t \rightarrow +\infty} \|\phi(t, X_0)\|$.

Here $\|Z\| = \sqrt{z_1^2 + z_2^2}$ if $Z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$.

5. (5 points) Consider the second order ODE depending on the parameter $\varepsilon > 0$:

$$x''(t) + \pi^2 x(t) = \cos((\pi + \varepsilon)t).$$

- (a) Find the general solution $x_\varepsilon(t)$ of the above equation depending on the parameter $\varepsilon > 0$.

- (b) Calculate $\lim_{\varepsilon \rightarrow 0} |x_\varepsilon(2)|$.

6. (5 points) Consider the second order ODE:

$$x''(t) = xe^{-x} - 3x \cos^2 x - 2x.$$

- (a) Find the equilibrium solutions of the above equation.

- (b) Determine the stability behaviour of the equilibrium solutions of the above equation.

7. (5 points) Find all the equilibrium solutions and determine its stability behaviour for the following equations:

(a) $x'(t) = -3x - x^3$,

(b) $x'(t) = 2x - x^2$,

(c) $x'(t) = -1 + x^2$.

8. (5 points) Suppose $\phi(t)$ satisfies: $\phi'(t) = \cos(1 + \phi^2) \sin(1 - \phi^2)$. If $|\phi(0)| \leq 1$, then show that $\phi(t)$ is defined for all $t \in \mathbb{R}$ and $|\phi(t)| \leq 1$ for all $t \in \mathbb{R}$.

9. (4 points) Suppose both ϕ and ψ satisfy the equation: $x'(t) = e^{1+\sin x(t)}$ for $|t| < \pi$. Show that:

$$|\phi(t) - \psi(t)| \leq |\phi(0) - \psi(0)| e^{te^2} \text{ for all } t \in (-\pi, \pi).$$

ex

Ordinary Differential Equations

Homework 1

Important

- Write your solutions neatly and submit it on 29 August (tutorials). Late submission will not be allowed.
- Simplify all your answers as much as possible and express answers in terms of fractions or constants such as $\sqrt{e}$ or $\ln(4)$ rather than decimals.
- Show all your work and explain your reasonings clearly! Copying will not be tolerated.

1. Find the general solution of the ODEs:

(a) $\frac{dx}{dt} = te^t.$

(b) $\frac{dx}{dt} = \frac{2x^4 + t^4}{tx^3}.$

(c) $\frac{dx}{dt} = \begin{cases} -t\sqrt{x} & \text{for } x \geq 0, \\ t\sqrt{-x} & \text{for } x \leq 0. \end{cases}$

2. Solve the following initial value problems:

$$\frac{dx}{dt} + a(t)x = t, \text{ with } x(0) = 1,$$

$$\text{where } a(t) = \begin{cases} 1 & \text{for } 0 \leq t \leq 2, \\ 3 & \text{for } t > 2. \end{cases}$$

Is the solution differentiable in $(0, +\infty)$.

3. Suppose $x(t)$ satisfies the ODE:

$$\dot{x}(t) = \alpha(t)x(t) + \beta(t),$$

where $\alpha, \beta : \mathbb{R} \rightarrow \mathbb{R}$ are continuous functions such that $\alpha(t) \leq -c < 0$ for some $c > 0$, and $\lim_{|t| \rightarrow +\infty} \beta(t) = 0$. Find $\lim_{t \rightarrow \infty} x(t)$.

4. Suppose $x(t)$ satisfies the ODE:

$$\frac{dx}{dt} - \frac{x}{2} = 2 \cos t.$$

Find the initial values $x(0) = x_0$ such that:

(a) $x(t) > 0$ as $t \rightarrow +\infty$.

(b) $x(t) < 0$ as $t \rightarrow +\infty$.

5. Solve the Bernoulli type equation

$$tx^2 \frac{dx}{dt} + x^3 = t \cos t.$$

6. Determine the constant a so that the following equations are exact, and then solve the resulting ODEs.

(a) $(2xt^3 + e^{at+x}) \frac{dx}{dt} + e^{at+x} + 3t^2x^2 = 0.$

(b) $\left(\frac{at+1}{x^3}\right) \frac{dx}{dt} + \frac{1}{x^2} + \frac{1}{t^2} = 0.$ ✓

7. Finding an integrating factor, or otherwise, solve the following ODEs:

(a) $(x^2 + t^2) \frac{dx}{dt} + 3t^2x + 2tx + x^3 = 0.$

(b) $(e^t \cos x + 2 \cos t) \frac{dx}{dt} + e^t \sin x - 2x \sin t = 0$, with $x(0) = \frac{\pi}{2}.$

(c) $(te^{tx} \cos(2t) - 3) \frac{dx}{dt} + xe^{tx} \cos(2t) - 2e^{tx} \sin(2t) + 2t = 0$, with $x(0) = 0.$

$\int \frac{N_x - M_y}{M} dx.$